

AZIZUR RAHMAN

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Research Interests

Deep Learning & Convolutional Neural Networks · Computer Vision & Medical Image Analysis · CNNTransformer Hybrid Architectures · Transfer Learning & Efficient Deep Learning · Explainable & Trustworthy AI · Multimodal Learning · Uncertainty Quantification & Robust Visual Learning

Research Profile

Computer Science researcher with an M.S. in Computer Science specialising in Artificial Intelligence (GPA: 3.97 / 4.00), with research experience spanning deep learning, convolutional neural networks, computer vision, medical imaging, Transformer architectures, transfer learning, multimodal learning, and explainable AI. My work focuses on developing robust, efficient, and interpretable deep learning architectures for visual recognition and medical image analysis, with particular interest in CNN-based representation learning, CNNTransformer hybrid models, transfer learning, and uncertainty-aware prediction. Application domains include gastrointestinal endoscopy risk stratification, Parkinson's disease detection, agricultural disease recognition, and housing affordability forecasting. My long-term research direction centres on building efficient, interpretable, and uncertainty-aware deep learning systems that are reliable enough for high-stakes real-world deployment particularly in medical imaging and clinical decision support.

Education

Master of Science in Computer Science Artificial Intelligence *Completed April 2026*
Indiana Wesleyan University, United States **GPA: 3.97 / 4.00**

Relevant coursework: Machine Learning Fundamentals, Model Development, Python Algorithms, Database Systems, Object-Oriented Programming, Cloud Computing.

Capstone Project: AI Integration Capstone designed and evaluated a deep learning pipeline for medical image classification, applying transfer learning, data augmentation, and model interpretability techniques for clinical decision support.

Bachelor of Science in Computer Science and Engineering *Completed February 2022*
Metropolitan University, Bangladesh **CGPA: 3.48 / 4.00**

Relevant coursework: Data Structures, Algorithms, Object-Oriented Programming, Machine Learning, Artificial Intelligence, Bioinformatics Computing, Discrete Mathematics.

Final Year Project: [Full-Stack E-Commerce Platform](#) built a production-grade online store using Django and React with JWT authentication, product comparison, shopping cart, and an integrated ML regression model for real-time laptop price prediction.

Honours & Awards

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- **Summa Cum Laude** Indiana Wesleyan University, April 2026 (GPA: 3.97 / 4.00)
 - **First Class** Bachelor of Science in Computer Science and Engineering, Metropolitan University, Bangladesh, February 2022 (CGPA: 3.48 / 4.00)
 - **Springer Nature Publication Acceptance** Paper accepted in *Applied Spatial Analysis and Policy* (Springer Nature, 2026); competitive peer-reviewed journal in spatial data science and urban analytics.

Publications

Beyond Binary: Four-Class Risk Stratification from Gastrointestinal Endoscopy Using Asymmetric-Cost Lightweight CNNTransformer Learning

Azizur Rahman, Nakib Uddin Ahmed, Mehjabin Ferdous, Kadirur Rahman Chowdhury
Medical & Biological Engineering & Computing (MBEC), Springer, 2026 [Under Review]

- Developed a four-class gastrointestinal lesion risk stratification framework using lightweight CNN and Transformer architectures (DenseNet-121, EfficientNet-B0, DeiT-Tiny).
- Mapped 21 of 23 HyperKvasir finding classes into four clinically actionable risk tiers; achieved macro F1 = 0.84 on HyperKvasir and 0.78 zero-shot on Kvasir-v2.
- Proposed an Asymmetric Endoscopy Loss (AEL) with a $5\times$ high-risk penalty; applied MC Dropout ($T = 30$, $\tau = 0.75$) for uncertainty-aware clinical referral decisions.
- Automated triage of 44.9% low-confidence cases with zero missed high-risk lesions.

AI-Driven Housing Affordability Forecasting in New York City: An NTA-Level Panel Analysis Using Ensemble Machine Learning

Azizur Rahman, Nakib Uddin Ahmed, Mehjabin Ferdous, Kadirur Rahman Chowdhury
Applied Spatial Analysis and Policy, Springer Nature, 2026 [Accepted]

- Built an explainable ensemble ML framework from 2,512 observations across 239 NYC Neighbourhood Tabulation Areas using 46 features (ACS, eviction records, Zillow ZORI).
- LightGBM achieved $R^2 = 0.926$ and RMSE = 0.035 under strict temporal cross-validation; applied SHAP for policy-level feature attribution.

MultiParkNet: A Ten-Stream Multimodal Deep Learning Framework with Cross-Modal Transformer Fusion and Dynamic Inter-Modal Gating for Early Detection of Parkinson's Disease

Nur Mohammad Oli, **Azizur Rahman**, A KM Rashid Been Aziz, *et al.*
Frontiers in Medicine, Precision Medicine, 2026 [Under Review]

- Co-developed a ten-stream multimodal framework integrating audio, motor drawing, structural MRI, ECG, and DaTSCAN modalities with cross-modal Transformer fusion.
- Achieved AUC-ROC = 0.9924, accuracy = 96.0%, and F1 = 0.9254 on the Parkinson Multi-Modal Dataset 2.0 (831 subjects).

CV-Mamba: A Leakage-Free Deep Learning Framework for Mango Leaf Disease Detection with MambaOut Transfer Learning

Shandipan Paul, Indrila Dey Traye, AKM Rashid Been Aziz, Abdul Kader, Fahim Muntashir, **Azizur Rahman**, Nakib Uddin Ahmed, Mithila Arman, Samuel Kim

Frontiers in Agronomy, Disease Management, 2026

[Under Review]

- Contributed to an eight-class mango leaf disease pipeline using MambaOut architectures with perceptual-hash deduplication and leakage-free partitioning.
- MambaOut-Small achieved 99.59% test accuracy and macro F1 = 0.9962.

Enhancing Online Recruitment Fraud Detection: A Comparative Analysis of Gradient Boosting and Transformer Architectures Under Severe Class Imbalance

Azizur Rahman, Nakib Uddin Ahmed

International Journal of Computer Applications, 2024

[Published]

- Compared BERT, RoBERTa, SVM, Random Forest, and XGBoost on 17,883 records with SMOTE; XGBoost reached 99.44% accuracy and F1 = 0.99.

Synergizing Lifestyle Behaviors and Metabolic Biomarkers: A Multi-Stage Ensemble Framework for Pre-Clinical Alzheimer's Screening

Avishek Roy Sparsho, **Azizur Rahman**, Joy Deep Roy, *et al.*

International Conference on Information Technology and Artificial Intelligence (ICITAI 2026), March 2026

[Accepted]

- Evaluated 18 ML algorithms on 2,149 patients using clinical, lifestyle, cognitive, and metabolic features; Bagging Classifier achieved 95.81% accuracy and AUC = 0.9625.

Professional Experience

Software Engineer Intern

March 2025 – July 2025

Next Generation Innovation L.L.C., New York, United States

- Led architectural design and deployment of scalable full-stack ecosystems using Django and FastAPI for service-oriented backends.
- Integrated predictive analytics and ML modules into enterprise software to enhance data-driven decision-making.
- Orchestrated cross-platform mobile strategies with React Native; optimised frontend performance via Next.js server-side rendering.
- Designed and managed databases across PostgreSQL, MySQL, MongoDB, and Firebase.

Software Engineer / SQA Engineer

January 2022 – December 2022

Inverse AI, Bangladesh

- Directed QA cycles and automated testing frameworks for system reliability and performance under load.

- Refined CI/CD pipelines and improved codebase maintainability through rigorous code reviews and defect analysis.

Selected Technical Projects

Breast Cancer Classifier *Python, Scikit-Learn, XGBoost, FastAPI* [\[GitHub\]](#)

- Trained four classifiers (Logistic Regression, SVM, Random Forest, XGBoost) on the Wisconsin Breast Cancer Diagnostic dataset (569 samples, 30 FNA features); tuned decision threshold to achieve $\geq 98\%$ recall on malignant cases, minimising false negatives in a screening context.

FaceAttend Face Recognition Attendance System *Python, PyTorch, ResNet-18, OpenCV, FastAPI* [\[GitHub\]](#)

- Multi-tenant biometric system with fine-tuned ResNet-18; novel two-layer liveness pipeline (texture analysis + blink detection) to mitigate photo/video spoofing.

DQN LunarLander Reinforcement Learning Agent *Python, PyTorch, Gymnasium, FastAPI* [\[GitHub\]](#)

- Implemented Double DQN to mitigate overestimation bias in the LunarLander-v3 environment; built a web UI for frame-by-frame agent visualisation and reward curves.

Real-Time Emotion Detector *Python, PyTorch, ResNet-18, OpenCV, FER2013* [\[GitHub\]](#)

- Fine-tuned ResNet-18 for seven-class facial emotion recognition from live webcam frames with real-time inference.

Technical Skills

Programming	Python, JavaScript, TypeScript, Java, C++, C
Machine Learning	Scikit-learn, XGBoost, LightGBM, TensorFlow, Pandas, NumPy, SHAP
Deep Learning	CNNs, Vision Transformers, Transformer Architectures, Transfer Learning, Multimodal Learning, MambaOut
Research Methods	Computer Vision, NLP, Explainable AI, Imbalanced Learning, Feature Engineering, Model Evaluation, Cross-Validation, Temporal Validation, Spatial Analysis, Forecasting
Medical AI	Medical Imaging, Clinical Decision Support, Risk Stratification, Uncertainty Quantification
Web & Mobile	Django, FastAPI, React.js, Next.js, React Native, Node.js, Express.js
Databases	PostgreSQL, MySQL, MongoDB, Firebase
Tools	Git, GitHub, Docker, Jupyter Notebook, REST APIs

Languages

English	Fluent
Bengali	Native
Hindi	Conversational

References

Dr. Santanu Das

Supervisor

Next Generation Innovation L.L.C., New York

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Lawrence Iweriebor

Professor, Computer Science

Indiana Wesleyan University, United States

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